MI DPU API

Version 0.1

REVISION HISTORY

Revision No.	Description	Date
0.1	• Initial release	01/10/2022

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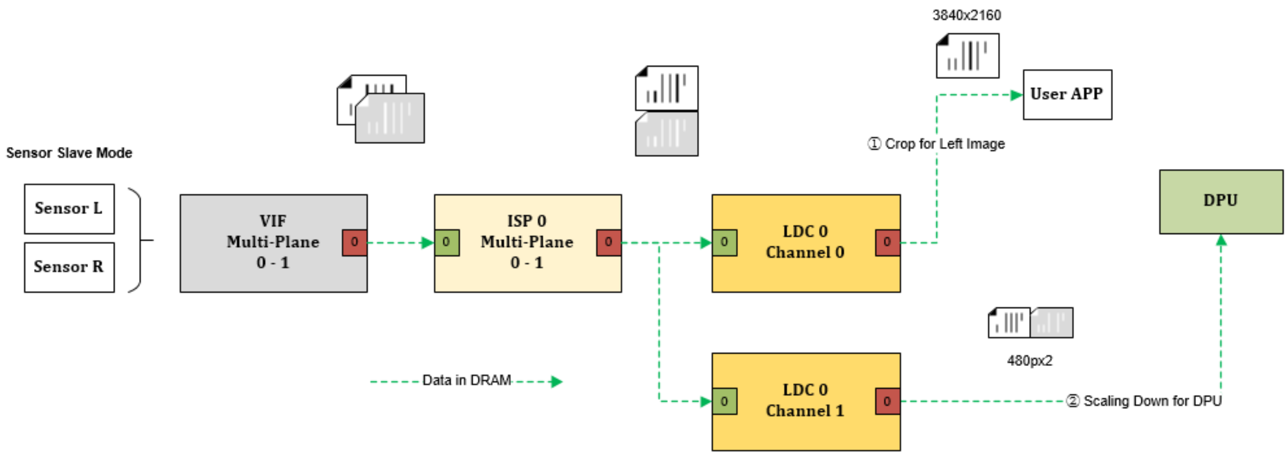
1. OVERVIEW

1.1. Module description

DPU module receive left and right frame, which are from two sensors, and output a buffer which is the same resolution as left frame with disparity info in it. This disparity info can be used to calculate the distance between the object and the sensor.

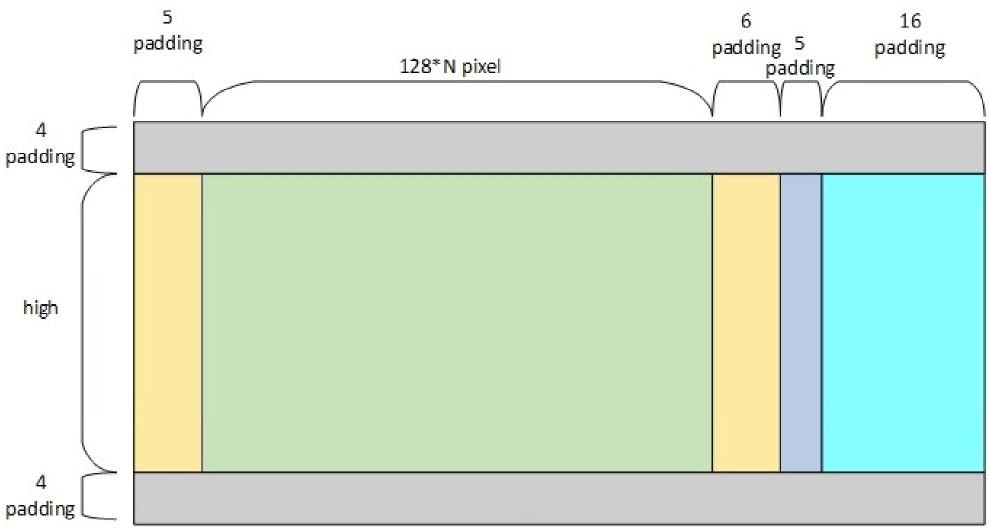
1.2. Flow chart



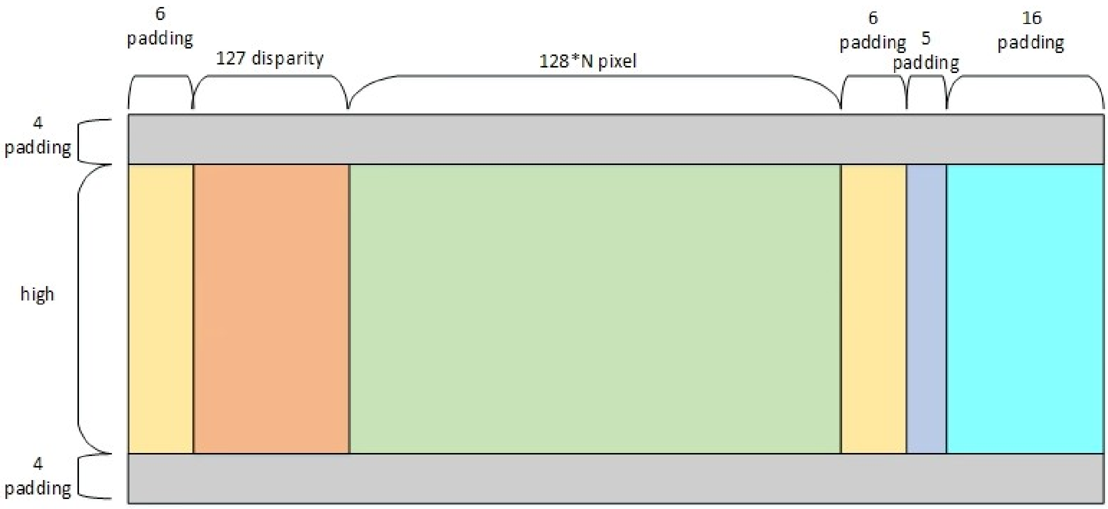
1.3. Keyword

DPU input:

Left buffer



Right buffer



The green block in the two pictures above, is the actual data from sensor, and they are needed 128 pixels align. The other color blocks are all their padding.

1. The yellow block which padding=5 in left picture, is acquired by algorithm. Its value is the same as the first column of green block. In the right part of the green block, the yellow block which padding=5 in left picture, is acquired by algorithm. The purple block which padding=5 is used to ensure 16 pixel align. The light blue block which padding=16, is required by algorithm. Their value are the same as last column of green block.
2. The right picture is about the same as left picture, except the orange block padding=127 pixels (Muffin only support disparity=64, so padding=127 should be changed to padding=63). To ensure 128 pixels align, yellow block padding=5+1=6. The yellow block and the orange block besides, have equivalent value of the first column of green block.

DPU input buffer, which described above, is output by the LDC module. The user should take no care of it. We provide cv tool for making a mapping table that used by LDC module.

You can know how to use cv tool from the document: <>

Device：
DPU Hardware device

Channel：
Time division multiplexed channel on device

Input image format：
Yuv420

Output image format：
disparity

Input resolution：
Muffin:
Only support 640 x 480

Output resolution：

Same as input resolution

2. API Reference

This function module provides the following user APIs。

API Name	Function
MI_DPU_CreateDevice	Create a DPU Device
MI_DPU_DestroyDevice	Destroy a DPU Device
MI_DPU_CreateChannel	Create a DPU Channel
MI_DPU_DestroyChannel	Destroy a DPU Channel
MI_DPU_StartChannel	Start DPU Channel
MI_DPU_StopChannel	Stop DPU Channel
MI_DPU_SetChnParam	Set DPU Channel parameter
MI_DPU_GetChnParam	Get DPU Channel parameter

2.1.1 MI_DPU_CreateDevice

➢ Description

Create a DPU device。

➢ Syntax

MI_S32 MI_DPU_CreateDevice(MI_DPU_DEV DevId , MI_DPU_DevAttr_t *pstDevAttr);

➢ Parameter

Parameter name	Description	Input/Output
DevId	DPU Device ID	Input
pstDevAttr	DPU Device attribute pointer, static attribute	Input

➢ Return value

- MI_SUCCESS(0) successful.
- Non-Zero: failed, see error code for details。

➢ Dependence

- Header: mi_dpu.h
- Library: libmi_dpu_user.a / libmi_dpu_user.so

※ Note

- Currently only DevId =0 is supported;

➢ Example

MI_DPU initialization and de-initialization

```
MI_DPU_DEV   DevId =0;
MI_DPU_CHANNEL   ChnId =0;
MI_DPU_DevAttr_t   stDevAttr = {0};
MI_DPU_ChnAttr_t  stChnAttr = {0};
```

```
stChnAttr.u16Width = 640;//设置 CHN window 宽度
stChnAttr.u16Height = 480;//设置 CHN window 高度
stChnAttr.u16MinDisparity = 0;//设置 DPU 起始 disparity
stChnAttr.u6MaxDisparity = 64;//设置 DPU 最大 disparity

 MI_DPU_CreateDevice(DevId, &stDevAttr);
 MI_DPU_CreateChannel(DevId, ChnId, &stChnAttr);
 MI_DPU_StartChannel(DevId, ChnId);

 /*destroy  channel*/
 MI_DPU_StopChannel(DevId, ChnId);
 MI_DPU_DestroyChannel(DevId,ChnId);
 MI_DPU_DestroyDevice(DevId)
```

➢ Related APIs
MI_DPU_DestroyDevice

2.1.2 MI_DPU_DestroyDevice

➢ Description
Destroy a DPU device。

➢ Syntax
MI_S32 MI_DPU_DestroyDevice(MI_DPU_DEV DevId);

➢ Parameter

Parameter name	Description	Input/Output
DevId	Device ID	Input

➢ Return value
- MI_SUCCESS(0) successful.
- Non-Zero: failed, see error code for details。

➢ Dependence
- Header: mi_dpu.h
- Library: libmi_dpu_user.a / libmi_dpu_user.so

※ Note
- Currently only DevId = 0 is Supported;

➢ Example
Set example of MI_DPU_CreateDevice

- Related APIs
  MI_DPU_CreateDevice

2.1.3 MI_DPU_CreateChannel

- Description
  Create a DPU Channel 。

- Syntax
  MI_S32 MI_DPU_CreateChannel(MI_DPU_DEV DevId , MI_DPU_CHANNEL ChnId，MI_DPU_ChnAttr_t *pstChAttr);

- Parameter

Parameter name	Description	Input/Output
DevId	DPU Device ID	Input
ChnId	Channel ID	Input
pstChAttr	Channel attribute	Input

- Return value
  - MI_SUCCESS(0) successful.
  - Non-Zero: failed, see error code for details。

- Dependence
  - Header: mi_dpu.h
  - Library: libmi_dpu_user.a / libmi_dpu_user.so

※ Note
  - Currently only ChnId = 0 is Supported；
  - Create Device firstly, and then create channel

- Example
  Set example of MI_DPU_CreateDevice

- Related APIs
  MI_DPU_DestroyChannel

2.1.4 MI_DPU_DestroyChannel

- Description
  Destroy a DPU Channel。

- Syntax
  MI_S32 MI_DPU_DestroyChannel(MI_DPU_DEV DevId , MI_DPU_CHANNEL ChnId);

➢ Parameter

Parameter name	Description	Input/Output
DevId	Device ID	Input
ChnId	Channel ID	Input

➢ Return value
- MI_SUCCESS(0) successful.
- Non-Zero: failed, see error code for details。

➢ Dependence
- Header: mi_dpu.h
- Library: libmi_dpu_user.a / libmi_dpu_user.so

※ Note
- Currently only ChnId = 0 is Supporte;
- Create Device firstly, and then Create channel

➢ Example

Set example of MI_DPU_CreateDevice

➢ Related APIs

MI_DPU_CreateChannel

2.1.5 MI_DPU_SetChnParam

➢ Description

Set DPU Channel Parameter.

➢ Syntax

MI_S32 MI_DPU_SetChnParam(MI_DPU_DEV DevId , MI_DPU_CHANNEL ChnId，MI_DPU_ChnParam _t *pstChnParam);

➢ Parameter

Parameter name	Description	Input/Output
DevId	Device ID	Input
ChnId	Channel ID	Input
pstChnParam	dynamic channel parameters	Input

➢ Return value
- MI_SUCCESS(0) successful.
- Non-Zero: failed, see error code for details。

➢ Dependence
- Header: mi_dpu.h
- Library: libmi_dpu_user.a / libmi_dpu_user.so

- Note
  - Currently only ChnId = 0 is Supported;
  - Create Device firstly,and then Create channel

➢ Example

```
MI_DPU_DEV   DevId =0;
MI_DPU_CHANNEL   ChnId =0;

MI_DPU_ChnParam_t  stChnParam = {0};

MI_DPU_GetChnParam(DevId , ChnId,& stChnParam);

stChnAttr.u16Width = 640;//设置 CHN window 宽度
stChnAttr.u16Height = 480;//设置 CHN window 高度
stChnAttr.u16MinDisparity = 0;//设置 DPU 起始 disparity
stChnAttr.u6MaxDisparity = 64;//设置 DPU 最大 disparity

MI_DPU_SetChnParam(DPUDevId , DPUChnId, & stChnParam);
```

➢ Related APIs
MI_DPU_GetChnParam

2.1.6 MI_DPU_GetChnParam

➢ Description
Get DPU Channel Parameter.

➢ Syntax
MI_S32 MI_DPU_GetChnParam(MI_DPU_DEV DevId , MI_DPU_CHANNEL ChnId，MI_DPU_ChnParam *pstChnParam);

➢ Parameter

Parameter name	Description	Input/Output
DevId	Device ID	Input
ChnId	Channel ID	Input
pstChnParam	dynamic channel parameters	Output

➢ Return value
- MI_SUCCESS(0) successful.
- Non-Zero: failed, see error code for details。

➢ Dependence
- Header: mi_dpu.h
- Library: libmi_dpu_user.a / libmi_dpu_user.so

※ Note

N/A

➢ Example

Set example of MI_DPU_SetChnParam

➢ Related APIs

MI_DPU_SetChnParam

2.1.7 MI_DPU_StartChannel

➢ Description

Start DPU Channel。

➢ Syntax

MI_S32 MI_DPU_StartChannel(MI_DPU_DEV DevId , MI_DPU_CHANNEL ChnId);

➢ Parameter

Parameter name	Description	Input/Output
DevId	Device ID	Input
ChnId	Channel ID	Input

➢ Return value

- MI_SUCCESS(0) successful.
- Non-Zero: failed, see error code for details。

➢ Dependence

- Header: mi_dpu.h
- Library: libmi_dpu_user.a / libmi_dpu_user.so

- Note
  - Only ChnId =0 is supported
  - Create Device firstly, and then Create channel

➢ Example

See Example of MI_DPU_CreateDevice

➢ Related APIs

MI_DPU_StopChannel

2.1.8 MI_DPU_StopChannel

➢ Description

Stop DPU Channel 。

➢ Syntax

MI_S32 MI_DPU_StopChannel(MI_DPU_DEV DevId , MI_DPU_CHANNEL ChnId);

➢ Parameter

Parameter name	Description	Input/Output
DevId	Device ID	Input
ChnId	Channel ID	Input

➢ Return value
- MI_SUCCESS(0) successful.
- Non-Zero: failed, see error code for details。

➢ Dependence
- Header: mi_dpu.h
- Library: libmi_dpu_user.a / libmi_dpu_user.so

※ Note
N/A

➢ Example
See example of MI_DPU_CreateDevice

➢ Related APIs
MI_DPU_StartChannel

3. data type

3.1. Data structure list

Video input related data type definition is as following：

MI_DPU_DEV	Define DPU Device type
MI_DPU_CHANNEL	Define DPU Channel type
MI_DPU_DevAttr_t	Define DPU Device attribute parameter
MI_DPU_ChnAttr_t	Define DPU Channel static attribute parameter。
MI_DPU_ChnParam_t	Define DPU Channel dynamic attribute parameter。

3.2. MI_DPU_DEV

➢ Description
Define DPU Device type.

➢ Definition
typedef MI_U32 MI_DPU_DEV

※ Note
N/A

➢ Related data type and interface
N/A

3.3. MI_DPU_CAHNNEL

➢ Description
Define DPU Channel type.

➢ Definition
```
typedef MI_U32 MI_DPU_CHANNEL
```

※ Note
N/A

➢ Related data type and interface
N/A

3.4. MI_DPU_DevAttr_t

➢ Description
Define DPU Device attribute parameter。

➢ Definition
```
typedef struct MI_DPU_DevAttr_s
{
        MI_U32 u32Reserved;
 } MI_DPU_DevAttr_t;
```

➢ Member

Member name	Description
u32Reserved	Reserved data

※ Note
N/A

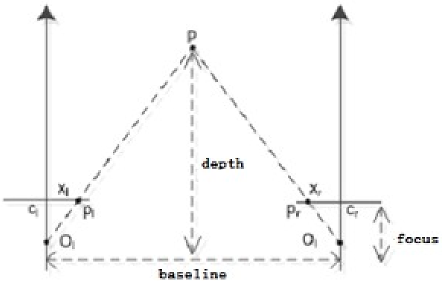
➢ Related data type and interface
N/A

3.5. MI_DPU_ChnAttr_t

➢ Description
Define DPU Channel static attribute parameter.

How to calculate MinDisparity and MaxDisparity:
The relationship of baseline, focus, disparity and depth is:

$$\frac{focal_{pixel} * baseline}{disparity} = depth$$



If the work range we need is: 0.6～4.1m
Suppose focus = 900pixels, baseline = 0.1m, thus
maxDisparity = 900 * 0.1 / 0.6 = 150;
minDisparity = 900 * 0.1 / 4.1 = 21.95;
or
minDisparity = maxDisparity – disparity = 150 – 128 = 22;

the conversion formula from pixel focus to millimeter focus is:

$$focus_{pixel} = \frac{focus_{mm}}{df}$$

And, df means the length of a pixel. The length of a pixel can be estimated according to the type of CMOS and its resolution. For example, a CMOS with 1/2.8 inch long, and its resolution is 2048 x 1536. We can get its image surface size is about 4mm x 3mm, thus

$$\mathrm{df} = \frac{width_{mm}}{width_{pixel}} \approx 2 * 10^{-3}\mathrm{mm}$$

if the millimeter focus is 3.6mm, we got that

$$focus_{pixel} = \frac{focus_{mm}}{df} = \frac{3.6}{2 * 10^{-3}} = 1800pixel$$

➢ Definition

```
typedef struct MI_DPU_ChnAttr_s
{
    MI_U16  u16Width;
    MI_U16  u16Height;
    MI_U16  u16MinDisparity;
    MI_U16  u16MaxDisparity;
} MI_DPU_ChnAttr_t;
```

➢ Member

Member name	Description
u16Width	input frame width
u16Height	input frame height
u16MinDisparity	minimum disparity
u16MaxDisparity	maximum disparity

※ Note
N/A

➢ Related data type and interface
N/A

3.6. MI_DPU_ChnParam_t

➢ Description
Define Dynamic parameters of DPU Channel

➢ Definition

```
typedef struct MI_DPU_ChnParam_s
{
        MI_U16  u16Width;
        MI_U16  u16Height;
        MI_U16  u16MinDisparity;
        MI_U16  u16MaxDisparity;
 } MI_DPU_ChnParam_t;
```

➢ Member

Member name	Description
u16Width	input frame width
u16Height	input frame height
u16MinDisparity	minimum disparity
u16MaxDisparity	maximum disparity

※ Note
N/A

➢ Related data type and interface
N/A

4. DPU ERROR CODE

API error codes are as the table shown below:

Error code	Macro definition	Description
0	MI_DPU_RET_SUCCESS	Success
0xA0282001	MI_ERR_DPU_MOD_NOT_INIT	Module is not initialized
0xA0282002	MI_ERR_DPU_DEV_ INVALID	Invalid device ID
0xA0282003	MI_ERR_DPU_CHN_INVALID	Invalid channel ID
0xA0282004	MI_ERR_DPU_TIMEOUT	DPU Device Running Timeout
0xA0282005	MI_ERR_DPU_NOMEM	Failed to allocate memory
0xA0282006	MI_ERR_DPU_NOT_SUPPORT	Not Supported
0xA0282007	MI_ERR_DPU_BUSY	Channel system is busy
0xA0282008	MI_ERR_DPU_NOT_READY	Device is not initialized
0xA0282009	MI_ERR_DPU_NULL_PTR	Null pointer parameter